

Web Development Course Outline (2 Months)

Duration: 2 Months

Classes: 5 per week

Class Duration: 1 Hour

Focus: JavaScript Advanced + Backend Basics (Node.js, Express, MongoDB) + Project

◆ MONTH 1 — JavaScript Advanced + Backend Fundamentals

WEEK 1 — JavaScript Advanced (Programming Foundations)

Class 1: JavaScript Revision & Execution Flow

- Revision of JS basics (variables, data types)
 - JavaScript execution (how JS runs)
 - `strict` mode
 - Best practices in JS coding
 - Real-life use cases
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Class 2: Arrays & Objects (Deep Understanding)

- What are arrays & objects
 - Array methods:
 - `push`, `pop`, `shift`, `unshift`
 - `map`, `filter`, `forEach`
 - Objects:
 - key-value pairs
 - accessing & updating data
 - Real-world examples
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Class 3: Functions Advanced

- Arrow functions
- Callback functions
- Scope (global, local, block)
- Hoisting (concept only)
- Practice problems

Class 4: Loops & Logic Building

- for loop
- while loop
- do-while
- Loop with arrays & objects
- Logic building exercises

Class 5: Error Handling & Debugging

- Common JS errors
- try / catch
- Debugging using browser DevTools
- Writing clean & readable JS
- Practice tasks

WEEK 2 — JavaScript & Browser APIs

Class 6: DOM Advanced

- DOM tree concept
- Selecting multiple elements
- Event bubbling & capturing
- `this` keyword (basic concept)
- Real UI examples

Class 7: Forms & Validation with JavaScript

- Form submit handling
- Prevent default behavior
- Input validation using JS
- Error messages & UI feedback
- Mini form project

Class 8: Local Storage & Session Storage

- What is browser storage?
- `localStorage` vs `sessionStorage`

- Storing & retrieving data
 - JSON stringify & parse
 - Small project: save user data
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Class 9: Asynchronous JavaScript

- Synchronous vs asynchronous
 - setTimeout & setInterval
 - Introduction to promises
 - Understanding async behavior (simple)
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Class 10: Fetch API (Intro)

- What is API?
 - JSON data format
 - Fetching data using fetch()
 - Handling response
 - Displaying API data on UI
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WEEK 3 — Backend Development Basics (Node.js + Express)

Class 11: Introduction to Backend & Node.js

- Frontend vs Backend
 - What is Node.js?
 - Why Node.js?
 - Installing Node.js
 - Running first Node program
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Class 12: NPM & Modules

- What is npm?
 - Installing packages
 - package.json
 - Import / export modules
 - Using built-in modules
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Class 13: Express.js Basics

- What is Express?
 - Creating server
 - Routing
 - Request & response
 - Sending data to browser
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Class 14: REST API Concepts

- What is REST?
 - HTTP methods:
 - GET
 - POST
 - PUT
 - DELETE
 - API structure
 - Testing APIs using browser / Postman (intro)
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Class 15: Mini Backend Project

- Simple Express API
 - Create routes
 - Handle data
 - Connect frontend with backend
 - Practice project
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◆ MONTH 2 (HALF MONTH) — Database + Final Project

WEEK 4 — Database & Full Stack Integration

Class 16: Introduction to Databases & MongoDB

- What is database?
 - SQL vs NoSQL
 - What is MongoDB?
 - Collections & documents
 - Installing MongoDB
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Class 17: MongoDB with Node.js

- Connecting MongoDB with Node
 - CRUD operations:
 - Create
 - Read
 - Update
 - Delete
 - Simple schema concept
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Class 18: Full Stack Flow

- Frontend → Backend → Database flow
 - Sending data from frontend
 - Receiving data in backend
 - Saving data in database
 - Error handling
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Class 19: Authentication Concept (Intro Level)

- What is authentication?
 - Login vs Signup flow
 - Password hashing concept
 - JWT concept (high-level)
 - Security best practices (intro)
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Class 20: Final Full Stack Project

Students will build a complete project such as:

- Contact Management App
- Student Record System
- Simple Login & Signup System
- CRUD Web Application

Includes:

- HTML, CSS, JavaScript
- Node.js & Express
- MongoDB
- Frontend + Backend integration